# SEC P vs NP log 2026-04-24

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-24 00:30 UTC

## SEC P vs NP — daily log 2026-04-24

*1 cycles recorded*

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### Chromatic Polynomial at -1 of Conflict Graphs Predicts #SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic graph theory (chromatic polynomials) × Number of satisfying assignments (#SAT)
- **Recorded**: 2026-04-24 00:03 UTC
- **Entry ID**: `166f507b54eb`

#### Statement

For every 3-CNF formula φ on n variables, let G_φ be its variable-conflict graph (with an edge between any two variables appearing together in a clause). Then the number of satisfying assignments of φ equals (-1)^n * χ_{G_φ}(-1), where χ_{G_φ} is the chromatic polynomial of G_φ evaluated at -1.

#### Rationale

The conflict graph captures pairwise co-occurrence of variables in clauses, a weak proxy for logical dependency. Chromatic polynomial evaluations at negative integers count acyclic orientations with sign, which may correspond to Boolean labelings avoiding contradictions. This surprising link could reveal combinatorial duality between graph colorings and solution-space structure.

#### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1608.04883v1] Approximating the Chromatic Polynomial - [2105.03304v2] Improved bounds for zeros of the chromatic polynomial on bounded degree graphs - [1901.00899v2] Recursion relations for chromatic coefficients for graphs and hypergraphs - [1901.05839v1] Hamiltonian chromatic number of block graphs - [1004.3702v97] A Polynomial time Algorithm for 3SAT

#### Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
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#### Judge reasoning

test_code_gen_failed